

API REFERENCE DOCUMENTATION

LMS Platform API

Production-ready REST API specification · v1.0 · 169 endpoints across 17 modules

Total Endpoints	Modules	Auth Required
169	17	140

Base URL

https://api.yourplatform.com



Global Conventions

- › All responses wrap data in: { success, data, message, meta }
- › Auth via: Authorization: Bearer <JWT>
- › Content-Type: application/json (unless file upload)
- › Errors follow RFC 7807 Problem Details format
- › Pagination params: ?page=1&limit=20
- › Rate limiting: 100 req/min (auth), 20 req/min (public)
- › Versioning: URL path /api/v1/...
- › CORS, HTTPS, and Helmet headers enforced on all routes



1. Authentication & Authorization

15 endpoints



REGISTRATION & LOGIN

POST

/api/v1/auth/register

Register a new user (student/instructor)

Body: `name`, `email`, `password`, `role`

POST

/api/v1/auth/login

Login and receive JWT access + refresh tokens

Body: `email`, `password`

POST

/api/v1/auth/logout  **AUTH**

Invalidate refresh token / blacklist JWT

POST

/api/v1/auth/refresh-token

Issue new access token using refresh token

Body: `refreshToken`

POST

/api/v1/auth/forgot-password

Send password reset email

Body: `email`

POST

/api/v1/auth/reset-password/:token

Reset password using token from email

Body: `newPassword`

POST

/api/v1/auth/verify-email/:token

Verify user email address

POST

/api/v1/auth/resend-verification  **AUTH**

Resend email verification link

OAuth / SOCIAL LOGIN

GET

/api/v1/auth/google

Redirect to Google OAuth consent screen

GET

/api/v1/auth/google/callback

Google OAuth callback — issue JWT

GET

/api/v1/auth/github

Redirect to GitHub OAuth consent screen

GET

/api/v1/auth/github/callback

GitHub OAuth callback — issue JWT

MFA

POST	/api/v1/auth/mfa/enable  AUTH Generate TOTP secret and QR code
POST	/api/v1/auth/mfa/verify  AUTH Confirm TOTP code to activate MFA Body: <code>totpCode</code>
POST	/api/v1/auth/mfa/disable  AUTH Disable MFA for the user Body: <code>totpCode</code>

 2. Users & Profiles 12 endpoints ▲		
PROFILE MANAGEMENT		
GET	/api/v1/users/me  AUTH	Get current authenticated user's profile
PUT	/api/v1/users/me  AUTH	Update profile (name, bio, headline, website, etc.)
DELETE	/api/v1/users/me  AUTH	Delete own account (GDPR soft-delete)
PUT	/api/v1/users/me/password  AUTH	Change password (requires current password)
POST	/api/v1/users/me/avatar  AUTH	Upload/replace profile avatar
GET	/api/v1/users/:userId/public	Get public profile of any user (instructor page)
ADMIN USER MANAGEMENT		
GET	/api/v1/admin/users  AUTH  ADMIN	List all users with filters (role, status, search)
GET	/api/v1/admin/users/:userId  AUTH  ADMIN	Get full user details
PUT	/api/v1/admin/users/:userId  AUTH  ADMIN	Update user role or status
DELETE	/api/v1/admin/users/:userId  AUTH  ADMIN	Hard-delete a user account
POST	/api/v1/admin/users/:userId/ban  AUTH  ADMIN	Ban/suspend a user account
POST	/api/v1/admin/users/:userId/unban  AUTH  ADMIN	Restore a banned user

 3. Courses 17 endpoints ▲		
COURSE DISCOVERY (PUBLIC)		
GET	/api/v1/courses List/search/filter courses (pagination, sort, category, price, rating, level)	
GET	/api/v1/courses/:courseId Get full course details (public landing page)	
GET	/api/v1/courses/featured Get featured/promoted courses	
GET	/api/v1/courses/trending Get trending courses by enrollment	
GET	/api/v1/courses/:courseId/reviews Get all reviews for a course (paginated)	
INSTRUCTOR COURSE MANAGEMENT		
POST	/api/v1/instructor/courses  AUTH  INSTRUCTOR Create a new course (draft)	
GET	/api/v1/instructor/courses  AUTH  INSTRUCTOR Get all courses created by the instructor	
GET	/api/v1/instructor/courses/:courseId  AUTH  INSTRUCTOR Get full editable details of a course	
PUT	/api/v1/instructor/courses/:courseId  AUTH  INSTRUCTOR Update course metadata (title, desc, price, thumbnail, etc.)	
DELETE	/api/v1/instructor/courses/:courseId  AUTH  INSTRUCTOR Delete/archive a course	
POST	/api/v1/instructor/courses/:courseId/submit  AUTH  INSTRUCTOR Submit course for admin review/approval	
POST	/api/v1/instructor/courses/:courseId/thumbnail  AUTH  INSTRUCTOR Upload course thumbnail image	
POST	/api/v1/instructor/courses/:courseId/promo-video  AUTH  INSTRUCTOR Upload course promotional video	
ADMIN COURSE MANAGEMENT		

GET	/api/v1/admin/courses  AUTH  ADMIN List all courses with filters (status, instructor, category)
POST	/api/v1/admin/courses/:courseId/approve  AUTH  ADMIN Approve course for publishing
POST	/api/v1/admin/courses/:courseId/reject  AUTH  ADMIN Reject course with feedback
DELETE	/api/v1/admin/courses/:courseId  AUTH  ADMIN Force-remove a course

 4. Curriculum (Sections & Lectures) 14 endpoints 	
SECTIONS	
GET	/api/v1/courses/:courseId/sections Get all sections of a course (ordered)
POST	/api/v1/instructor/courses/:courseId/sections  AUTH  INSTRUCTOR Add a section to the course
PUT	/api/v1/instructor/courses/:courseId/sections/:sectionId  AUTH  INSTRUCTOR Update section title/order
DELETE	/api/v1/instructor/courses/:courseId/sections/:sectionId  AUTH  INSTRUCTOR Delete a section (and its lectures)
PUT	/api/v1/instructor/courses/:courseId/sections/reorder  AUTH  INSTRUCTOR Reorder sections via ordered array of IDs
LECTURES	
POST	/api/v1/instructor/sections/:sectionId/lectures  AUTH  INSTRUCTOR Create a new lecture (video, article, quiz)
GET	/api/v1/sections/:sectionId/lectures/:lectureId  AUTH Get a lecture (checks enrollment for protected content)
PUT	/api/v1/instructor/sections/:sectionId/lectures/:lectureId  AUTH  INSTRUCTOR Update lecture metadata
DELETE	/api/v1/instructor/sections/:sectionId/lectures/:lectureId  AUTH  INSTRUCTOR Delete a lecture
PUT	/api/v1/instructor/sections/:sectionId/lectures/reorder  AUTH  INSTRUCTOR Reorder lectures in a section
POST	/api/v1/instructor/lectures/:lectureId/video  AUTH  INSTRUCTOR Upload/replace lecture video (triggers transcoding pipeline)
GET	/api/v1/instructor/lectures/:lectureId/video/status  AUTH  INSTRUCTOR Poll video transcoding status

POST	/api/v1/instructor/lectures/:lectureId/attachment  AUTH  INSTRUCTOR Upload downloadable resource/attachment
DELETE	/api/v1/instructor/lectures/:lectureId/attachment/:attachmentId  AUTH  INSTRUCTOR Delete an attachment



5. Enrollment & Student Access 9 endpoints



ENROLLMENT

POST	/api/v1/enrollments  AUTH Enroll in a course (free or post-payment) Body: <code>courseId</code>
GET	/api/v1/enrollments/me  AUTH Get all courses the student is enrolled in
GET	/api/v1/enrollments/me/:courseId  AUTH Check enrollment status for a course
GET	/api/v1/instructor/courses/:courseId/students  AUTH  INSTRUCTOR Get all enrolled students for a course

PROGRESS TRACKING

GET	/api/v1/progress/:courseId  AUTH Get full progress data for a course
POST	/api/v1/progress/:courseId/lectures/:lectureId/complete  AUTH Mark a lecture as completed
DELETE	/api/v1/progress/:courseId/lectures/:lectureId/complete  AUTH Unmark a lecture as completed
PUT	/api/v1/progress/:courseId/lectures/:lectureId/position  AUTH Save video playback position (resume) Body: <code>positionSeconds</code>
GET	/api/v1/progress/:courseId/certificate/eligibility  AUTH Check if student meets completion threshold for certificate

 6. Payments & Billing 18 endpoints ▲		
CHECKOUT		
POST	/api/v1/payments/checkout  AUTH	Create a checkout session (Stripe) Body: <code>courseId</code> , <code>couponCode?</code>
GET	/api/v1/payments/checkout/:sessionId  AUTH	Get checkout session status (poll after redirect)
POST	/api/v1/payments/webhooks/stripe	Stripe webhook endpoint (payment confirmation, refund, disputes)
ORDERS & HISTORY		
GET	/api/v1/payments/orders  AUTH	Get user's order history
GET	/api/v1/payments/orders/:orderId  AUTH	Get single order details + invoice
POST	/api/v1/payments/orders/:orderId/refund  AUTH	Request a refund (within refund window)
GET	/api/v1/payments/orders/:orderId/invoice  AUTH	Download order invoice as PDF
COUPONS & DISCOUNTS		
POST	/api/v1/coupons/validate  AUTH	Validate a coupon code and return discount info Body: <code>code</code> , <code>courseId</code>
POST	/api/v1/admin/coupons  AUTH  ADMIN	Create a coupon
GET	/api/v1/admin/coupons  AUTH  ADMIN	List all coupons with usage stats
PUT	/api/v1/admin/coupons/:couponId  AUTH  ADMIN	Update/deactivate a coupon
DELETE	/api/v1/admin/coupons/:couponId  AUTH  ADMIN	Delete a coupon
INSTRUCTOR PAYOUTS		
GET	/api/v1/instructor/earnings  AUTH  INSTRUCTOR	Get earnings summary (total, pending, paid)

GET	/api/v1/instructor/earnings/history	🔒 AUTH	👑 INSTRUCTOR	Detailed earnings transaction history
POST	/api/v1/instructor/payouts/request	🔒 AUTH	👑 INSTRUCTOR	Request a payout to bank/PayPal
GET	/api/v1/instructor/payouts	🔒 AUTH	👑 INSTRUCTOR	Get payout history and status
POST	/api/v1/admin/payouts/:payoutId/approve	🔒 AUTH	👑 ADMIN	Approve a payout request
POST	/api/v1/admin/payouts/:payoutId/reject	🔒 AUTH	👑 ADMIN	Reject a payout request



7. Reviews & Ratings

8 endpoints



REVIEWS

POST	/api/v1/courses/:courseId/reviews	🔒 AUTH		Submit a review (enrolled users only) Body: <code>rating</code> (1-5), <code>comment</code>
PUT	/api/v1/courses/:courseId/reviews/:reviewId	🔒 AUTH		Edit own review
DELETE	/api/v1/courses/:courseId/reviews/:reviewId	🔒 AUTH		Delete own review
POST	/api/v1/courses/:courseId/reviews/:reviewId/helpful	🔒 AUTH		Mark a review as helpful
POST	/api/v1/instructor/courses/:courseId/reviews/:reviewId/respond	🔒 AUTH	👑 INSTRUCTOR	Instructor response to a review
POST	/api/v1/courses/:courseId/reviews/:reviewId/report	🔒 AUTH		Flag a review as inappropriate
GET	/api/v1/admin/reviews/reported	🔒 AUTH	👑 ADMIN	List all flagged reviews for moderation
DELETE	/api/v1/admin/reviews/:reviewId	🔒 AUTH	👑 ADMIN	Admin-remove an inappropriate review



8. Categories & Tags

8 endpoints



CATEGORIES

GET

/api/v1/categories

Get all categories and subcategories (tree)

GET

/api/v1/categories/:categoryId/courses

Get courses by category (paginated)

POST

/api/v1/admin/categories AUTH ADMIN

Create a category/subcategory

PUT

/api/v1/admin/categories/:categoryId AUTH ADMIN

Update category name/slug/parent

DELETE

/api/v1/admin/categories/:categoryId AUTH ADMIN

Delete a category (reassign or orphan courses)

TAGS

GET

/api/v1/tags

List all tags (with optional search)

POST

/api/v1/admin/tags AUTH ADMIN

Create a tag

DELETE

/api/v1/admin/tags/:tagId AUTH ADMIN

Delete a tag



9. Search

3 endpoints



SEARCH ENDPOINTS

GET

/api/v1/search

Full-text search across courses, instructors, categories

GET

/api/v1/search/suggestions

Autocomplete/typeahead suggestions

Body: `query` param: `q`

GET

/api/v1/search/filters

Available filter options for current query results



10. Quizzes & Assignments

12 endpoints



QUIZZES

POST

/api/v1/instructor/lectures/:lectureId/quiz

AUTH

INSTRUCTOR

Create quiz for a lecture

PUT

/api/v1/instructor/quizzes/:quizId

AUTH

INSTRUCTOR

Update quiz (title, pass score, time limit)

POST

/api/v1/instructor/quizzes/:quizId/questions

AUTH

INSTRUCTOR

Add questions to a quiz

PUT

/api/v1/instructor/quizzes/:quizId/questions/:questionId

AUTH

INSTRUCTOR

Update a question

DELETE

/api/v1/instructor/quizzes/:quizId/questions/:questionId

AUTH

INSTRUCTOR

Delete a question

POST

/api/v1/quizzes/:quizId/attempt

AUTH

Submit a quiz attempt

Body: `answers[ ]`

GET

/api/v1/quizzes/:quizId/attempts

AUTH

Get student's attempt history for a quiz

GET

/api/v1/quizzes/:quizId/attempts/:attemptId

AUTH

Get detailed result of a specific attempt

ASSIGNMENTS

POST

/api/v1/instructor/lectures/:lectureId/assignment

AUTH

INSTRUCTOR

Create an assignment

POST

/api/v1/assignments/:assignmentId/submit

AUTH

Student submits assignment (file/text)

GET

/api/v1/instructor/assignments/:assignmentId/submissions

AUTH

INSTRUCTOR

Get all student submissions

PUT

**/api/v1/instructor/assignments/:assignmentId/
submissions/:submissionId/grade**

AUTH

INSTRUCTOR

Grade a submission with feedback



11. Certificates 5 endpoints



CERTIFICATE MANAGEMENT

POST	/api/v1/certificates/generate/:courseId	🔒 AUTH
Generate certificate upon course completion		
GET	/api/v1/certificates/me	🔒 AUTH
Get all certificates earned by the user		
GET	/api/v1/certificates/:certificateId	
Get certificate details (public for verification)		
GET	/api/v1/certificates/:certificateId/download	🔒 AUTH
Download certificate as PDF		
GET	/api/v1/certificates/verify/:verificationCode	
Verify authenticity of a certificate (public)		

 12. Wishlist & Cart 9 endpoints		
WISHLIST		
GET	/api/v1/wishlist  AUTH	Get user's wishlist
POST	/api/v1/wishlist  AUTH	Add course to wishlist Body: <code>courseId</code>
DELETE	/api/v1/wishlist/:courseId  AUTH	Remove course from wishlist
CART		
GET	/api/v1/cart  AUTH	Get current cart contents with pricing summary
POST	/api/v1/cart  AUTH	Add a course to cart Body: <code>courseId</code>
DELETE	/api/v1/cart/:courseId  AUTH	Remove a course from cart
DELETE	/api/v1/cart  AUTH	Clear entire cart
POST	/api/v1/cart/coupon  AUTH	Apply a coupon to cart Body: <code>code</code>
DELETE	/api/v1/cart/coupon  AUTH	Remove coupon from cart

 13. Notifications 6 endpoints ▲		
NOTIFICATIONS		
GET	/api/v1/notifications  AUTH	Get user notifications (paginated)
PUT	/api/v1/notifications/:notificationId/read  AUTH	Mark a notification as read
PUT	/api/v1/notifications/read-all  AUTH	Mark all notifications as read
DELETE	/api/v1/notifications/:notificationId  AUTH	Delete a notification
GET	/api/v1/notifications/preferences  AUTH	Get notification preferences
PUT	/api/v1/notifications/preferences  AUTH	Update notification preferences (email, push, in-app)



14. Q&A / Discussion

9 endpoints



QUESTIONS & ANSWERS

GET

/api/v1/courses/:courseId/questions AUTH

Get all Q&A for a course (paginated, filterable)

POST

/api/v1/courses/:courseId/questions AUTH

Post a question (enrolled students)

PUT

/api/v1/questions/:questionId AUTH

Edit own question

DELETE

/api/v1/questions/:questionId AUTH

Delete own question

POST

/api/v1/questions/:questionId/answers AUTH

Answer a question (instructor or student)

PUT

/api/v1/questions/:questionId/answers/:answerId AUTH

Edit own answer

DELETE

/api/v1/questions/:questionId/answers/:answerId AUTH

Delete own answer

POST

/api/v1/questions/:questionId/answers/:answerId/upvote AUTH

Upvote an answer

POST

/api/v1/questions/:questionId/answers/:answerId/mark-correct

AUTH

INSTRUCTOR

Instructor marks an answer as correct



15. Analytics & Reporting

10 endpoints



INSTRUCTOR ANALYTICS

GET

/api/v1/instructor/analytics/overview

AUTH



INSTRUCTOR

Summary: total students, revenue, avg rating

GET

/api/v1/instructor/analytics/revenue

AUTH



INSTRUCTOR

Revenue breakdown by course, date range

GET

/api/v1/instructor/analytics/enrollments

AUTH



INSTRUCTOR

Enrollment trends over time

GET

/api/v1/instructor/analytics/courses/:courseId

AUTH



INSTRUCTOR

Per-course: completion rate, drop-off per lecture, avg quiz score

GET

/api/v1/instructor/analytics/reviews

AUTH



INSTRUCTOR

Review sentiment trends

ADMIN ANALYTICS

GET

/api/v1/admin/analytics/overview

AUTH



ADMIN

Platform KPIs: MAU, GMV, total courses, instructors

GET

/api/v1/admin/analytics/revenue

AUTH



ADMIN

Platform-wide revenue and payout reports

GET

/api/v1/admin/analytics/users

AUTH



ADMIN

User growth, churn, and retention metrics

GET

/api/v1/admin/analytics/courses

AUTH



ADMIN

Course performance across the platform

POST

/api/v1/admin/analytics/reports/export

AUTH



ADMIN

Export analytics data as CSV/Excel

 16. Media / File Management 8 endpoints		▲
UPLOADS		
POST	/api/v1/media/upload/image  AUTH Upload image (course thumbnail, avatar) → returns CDN URL	
POST	/api/v1/media/upload/video/initiate  AUTH  INSTRUCTOR Initiate a multipart video upload → returns upload ID and parts	
PUT	/api/v1/media/upload/video/:uploadId/part/:partNumber  AUTH  INSTRUCTOR Upload individual video part (chunked)	
POST	/api/v1/media/upload/video/:uploadId/complete  AUTH  INSTRUCTOR Complete multipart upload → triggers transcoding	
DELETE	/api/v1/media/upload/video/:uploadId/abort  AUTH  INSTRUCTOR Abort and clean up a multipart upload	
GET	/api/v1/media/video/:videoId/stream  AUTH Get signed HLS/DASH streaming URL for a lecture video	
GET	/api/v1/media/video/:videoId/captions  AUTH Get auto-generated or uploaded subtitles/captions	
POST	/api/v1/media/video/:videoId/captions  AUTH  INSTRUCTOR Upload a subtitle/caption file (VTT/SRT)	

 17. System & Health 6 endpoints ▲	
SYSTEM	
GET	/api/v1/health Basic health check (liveness probe)
GET	/api/v1/health/ready Readiness probe (checks DB, cache, storage)
GET	/api/v1/health/metrics 🔥 INTERNAL Expose Prometheus metrics (internal only)
GET	/api/v1/admin/settings 🔒 AUTH 👑 ADMIN Get platform settings (maintenance mode, feature flags)
PUT	/api/v1/admin/settings 🔒 AUTH 👑 ADMIN Update platform settings
GET	/api/v1/admin/audit-logs 🔒 AUTH 👑 ADMIN Get audit trail of admin actions

LMS API Specification v1.0 · Generated for Production Use · All endpoints should be documented further in OpenAPI/Swagger format